

EXL

## Environment Service Voice Agent

### Team – A

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## The Problem

# Manual Operations At Scale

This Agent providers face mounting pressure from rising call volumes, missed pickups, and inefficient manual processes — driving up costs and eroding customer trust.

01

### Missed Pickups

No automated notification or resolution path for customers.

02

### Missed Pickups

High call center workload with long wait times and inconsistent responses.

03

### Missed Pickups

Customers lack visibility into service schedules or delays.

04

### Missed Pickups

Inefficient rescheduling and escalation processes inflate overhead.



## SOLUTION OVERVIEW

# The AI Voice Agent



An intelligent conversational AI Agent that understands natural language, automates interactions, and delivers faster, accurate and seamless service experiences

**Authenticate Users**

Secure identity verification with multi-setup login and validation.

**Handle Service Requests**

Raise pickup requests and resolve issues quickly and accurately.

**Provide Status Update**

Real-time service information and proactive updates.

**Escalate when needed**

Complex issues routed seamlessly to human agents.

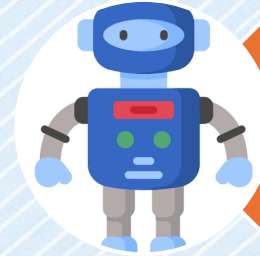
# SYSTEM OBJECTIVE

Six core goal drives the design of the AI voice agent – balancing user experience with operational efficiency.



## 24 X 7 Support

Always-on availability, eliminating wait times and after-hours gaps.



## Automate Requests

End-to-end automation for routine service interactions,



## Reduce Costs

Lower call center volume and manual processing overhead.



## Improve CX

Faster resolution and consistent responses boost satisfaction.



## Operational Efficiency

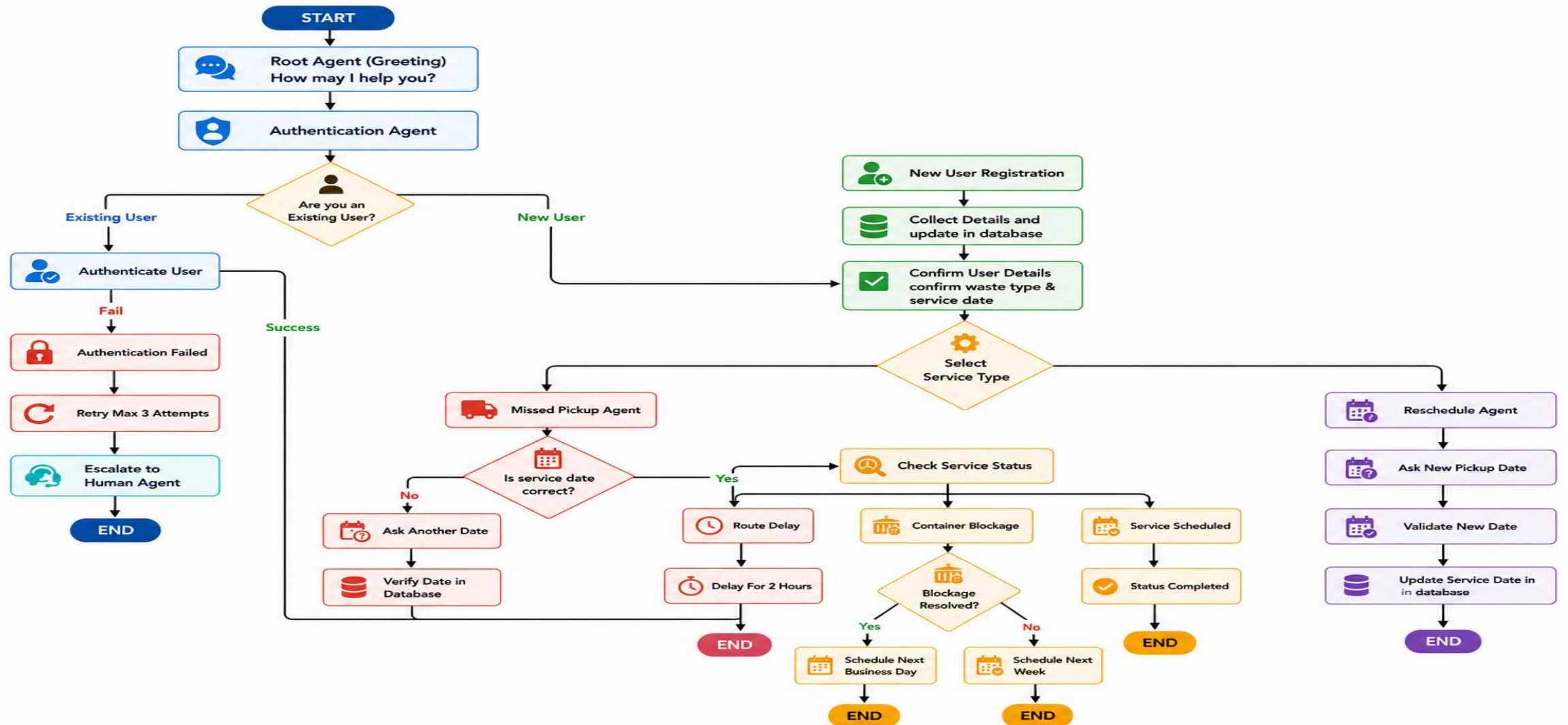
Streamlined workflows reduce agent workload and errors.



## Real-Time Tracking

Live service status updates delivered directly to customers

# AGENT WORKFLOW



# Authentication & Registration Flows

1

## Existing User Authentication

- User provides identifying information
- System verifies credentials against database
- Success → Proceed to service selection
- Failure → Up to 3 retry attempts
- After 3 failures → Escalate to human agent

2

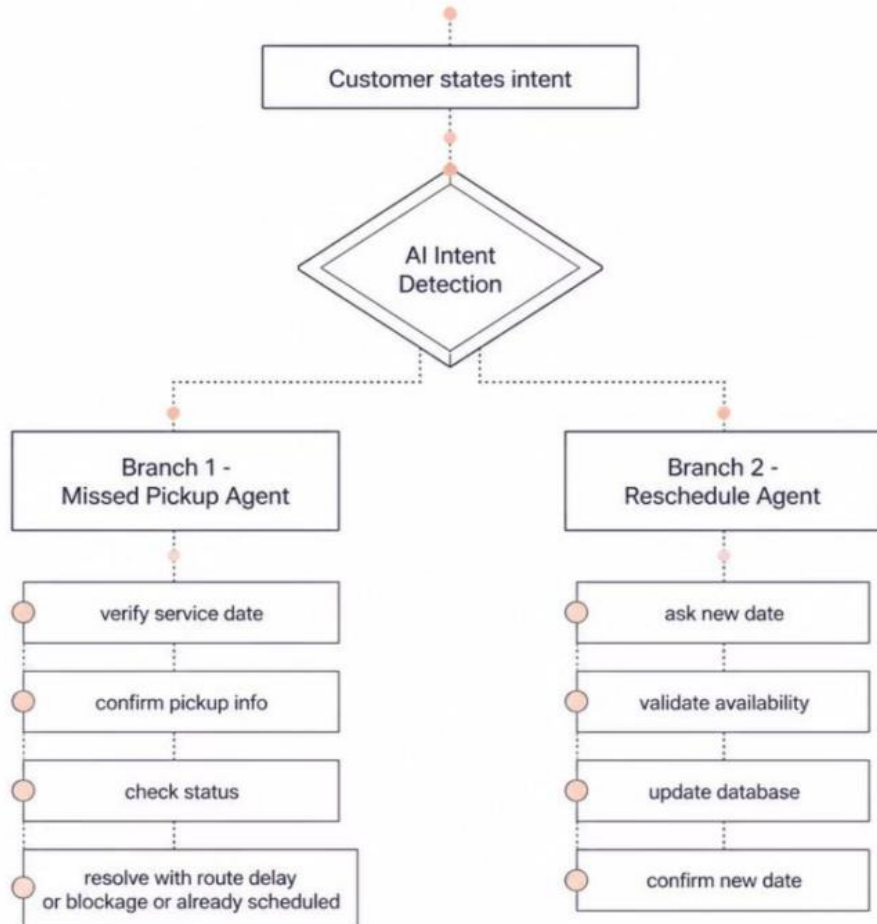
## New User Authentication

- Initiate new user registration - Collect name, address, and contact details.
- Update backend customer database.
- Confirm service type and coverage area.
- Confirm and schedule first pickup date

Registration data syncs in real time with the waste management CRM.

# Service Selection & Missed Pickup Resolution

The AI detects customer intent through natural language and routes the call to the appropriate specialized agent.



## Route Delay

Inform customer of ~2-hour delay.  
Request closed automatically.

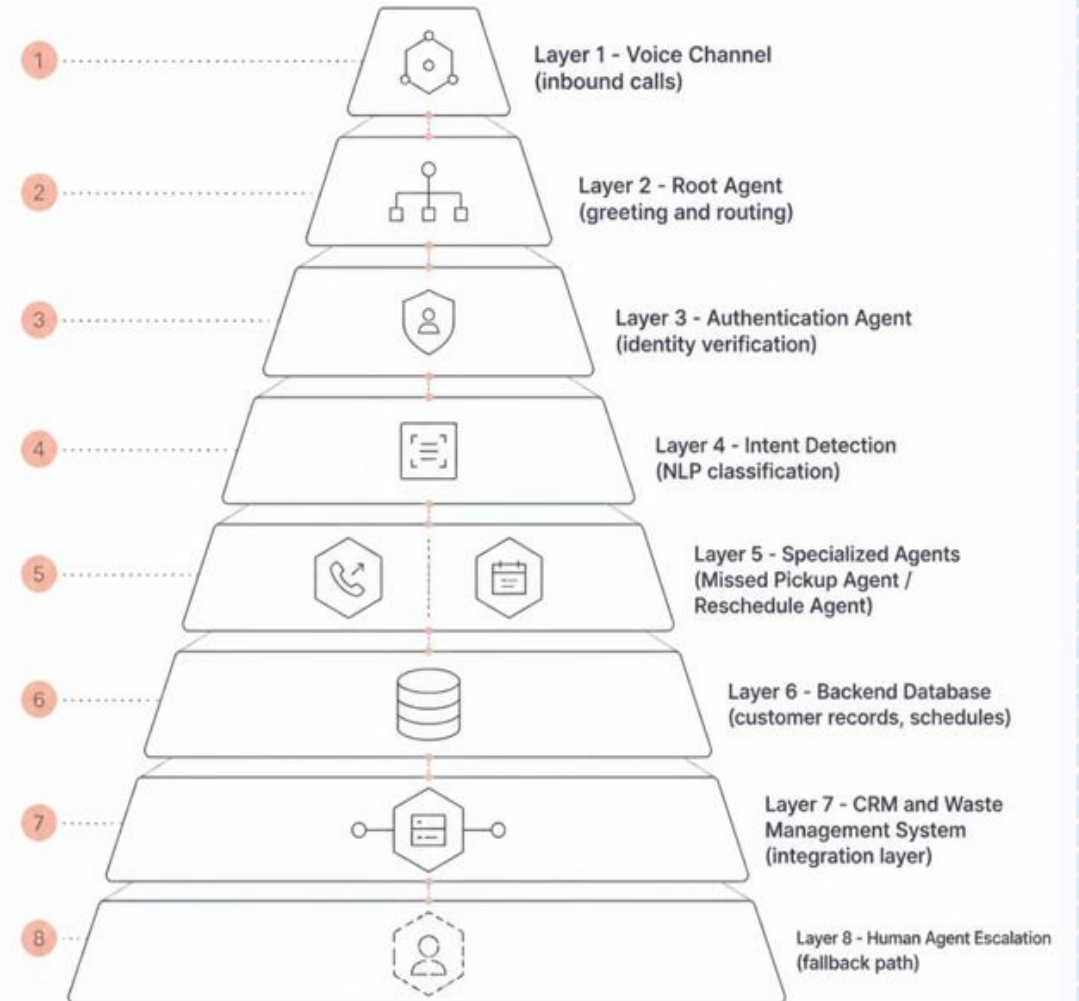
## Container Blockage

Resolved → Next business day.  
Unresolved → Schedule next week.

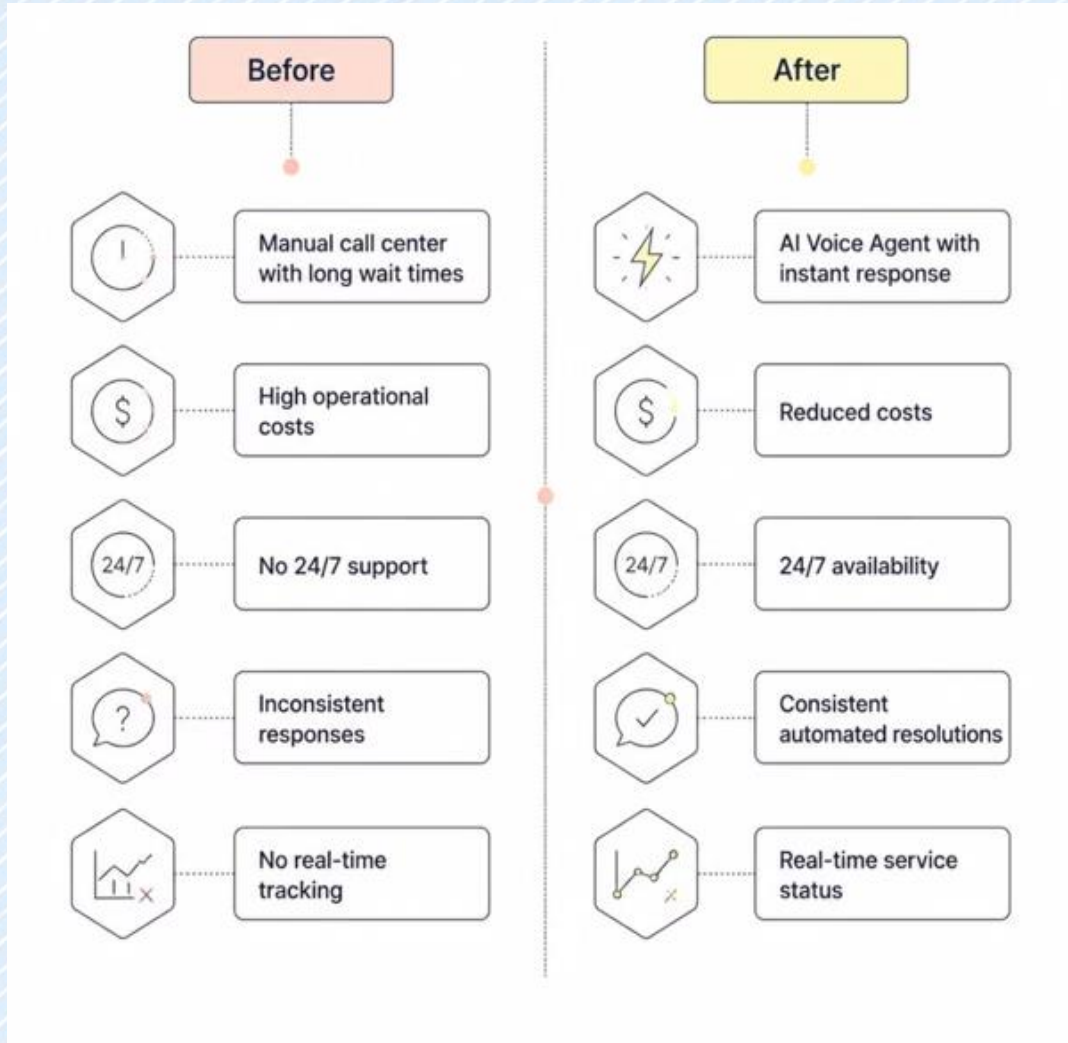
## Already Scheduled

Notify customer of existing pickup date.  
No action needed.

# Reschedule Agent & Technical Architecture



## BUSINESS IMPACT & BENEFITS



**24x7  
Availability**

No after-hours coverage gaps

**Reduced  
Call Volume**

Automation handles routine requests

**Faster  
Resolution**

Instant responses vs. manual processing

**Lower Costs**

Consistent, reliable customer experience

## Edge Cases



Invalid Address



Invalid Service Date



Authentication Failure



Service Status = Completed / Delayed / Scheduled



Container Block



Route Delay



No Active Service



Backend / API Failure



New Customer Not Found

## Fallbacks



Retry Address



Retry Service Date



Three Authentication Attempts



Escalation to Live Agent



Friendly Error Message on API Failure



Graceful End when No Active Service Exists



Switch to New User Registration When Customer is not found

**THANK YOU**